

Numerical and Machine Learning Methods for the Diabetes Glucose Tolerance Test

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Abstract—This report evaluates computational solutions for Randall’s 2-ODE model to simulate a 12-hour Oral Glucose Tolerance Test (OGTT) across four metabolic scenarios. To address the system’s non-linear feedback loops, we implement and contrast three strategies: a custom fixed-step RK4 method, an adaptive Dormand-Prince (RK45) scheme, and a Physics-Informed Neural Network (PINN). Benchmarks show that while fixed-step integration introduces discretization errors during sudden post-infusion spikes, the adaptive RK45 scheme efficiently balances precision with computational speed. Furthermore, the PINN framework successfully embeds physiological constraints to capture metabolic trends without grid dependencies, demonstrating strong potential for continuous glucose monitoring (CGM) systems.

Index Terms—Oral Glucose Tolerance Test, Randall Model, Runge-Kutta, Physics-Informed Neural Networks, Biomedical Simulation.

I. INTRODUCTION

The Oral Glucose Tolerance Test (OGTT) is a critical diagnostic tool for evaluating pancreatic responsiveness, yet discrete blood draws fail to capture continuous physiological feedback loops. To bridge this gap, we implement Randall’s 2-ODE model to simulate extracellular glucose $G(t)$ and insulin $I(t)$ dynamics over 12 hours. We benchmark three computational approaches to handle the system’s non-linear, multi-scale nature: a custom fixed-step RK4 solver, an adaptive Dormand-Prince (RK45) scheme, and a Physics-Informed Neural Network (PINN). Our work analyzes the core trade-offs between classical numerical accuracy and modern learning-based techniques for real-time metabolic monitoring.

II. LITERATURE REVIEW

This survey covers three pillars relevant to the project: ODE modelling of glucose–insulin dynamics, physics-informed and neural ODE methods for metabolic systems, and benchmarking of numerical integration schemes.

A. Coupled ODE Models of Glucose–Insulin Dynamics

Kumnungkit et al. [1] proposed a minimal glucose–insulin ODE model incorporating food intake, achieving $R^2 > 0.99$ against both minimal and maximal reference models. Their work confirms that low-dimensional coupled ODE systems can capture clinically relevant dynamics without the identifiability problems of high-dimensional compartmental models — a philosophy shared by the two-ODE system adopted in this project, where $G(t)$ and $I(t)$ are the sole state variables and the insulin effect appears as the nonlinear term $-G_g I G$.

Aliffi et al. [2] fitted stochastic ODE systems to CGM data from eight type-2 diabetic subjects using Particle Swarm Optimisation and information-theoretic model selection. They demonstrate the limitations of the classical three-variable Schiesser (Randall) 2-ODE Model formulation for OGTT data and support simpler two-state representations, consistent with the model used here.

B. Physics-Informed and Neural ODE Methods

Wang et al. [3] introduced PIGNN, embedding glucose–insulin ODE residuals directly into the network loss:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}}, \quad (1)$$

achieving 0.726 ± 0.126 mmol/L accuracy on 52 subjects and a 12.33% improvement over data-driven baselines with only 48 training samples.

Pathirana et al. [4] extended PINNs for glucose–insulin digital twins using Federated Learning, reporting a mean ODE residual of 0.033 and prediction errors within 10 mg/dL, demonstrating that physics constraints substantially improve robustness under sparse clinical data.

Kainth et al. [5] introduced PI-NODE-SR, combining a Heun solver with scale-aware residual normalisation to stabilise Neural ODE training on stiff biophysical systems — directly relevant to the fast post-load glucose spike followed by the slow insulin recovery phase characteristic of the OGTT.

C. Benchmarking of Numerical Integration Schemes

Städter et al. [6] benchmarked ODE solvers across biological models from the BioModels database, showing that adaptive multi-step methods consistently outperform fixed-step Runge–Kutta schemes on stiff systems in both accuracy and wall-clock time. These findings motivate the fixed-step RK4 versus adaptive RK45 comparison conducted in this project.

III. MATHEMATICAL MODEL & SIMULATION CASES

The Schiesser two-compartment ODE model [7] simulates the Oral Glucose Tolerance Test (OGTT) by tracking extracellular glucose $G(t)$ and insulin $I(t)$ over 12 hours. This model is based on the pioneering work of Randall [8]. The glucose balance is governed by:

$$C_g \frac{dG}{dt} = \begin{cases} Q + I_n - G_g I G - D_d G, & G < G_k, \\ Q + I_n - G_g I G - D_d G - \mu(G - G_k), & G \geq G_k, \end{cases} \quad (2)$$

where Q is basal hepatic production, I_n is the 30-minute oral glucose infusion ($G_t = 80,000$ mg), $-G_g I G$ is insulin-mediated uptake, $-D_d G$ is insulin-independent metabolism, and $-\mu(G - G_k)$ is renal excretion active only above the threshold $G_k = 250$ mg/Ll. Insulin evolves according to:

$$C_i \frac{dI}{dt} = \begin{cases} -A_a I, & G < G_0, \\ -A_a I + B_b(G - G_0), & G \geq G_0, \end{cases} \quad (3)$$

where $-A_a I$ is first-order clearance and $B_b(G - G_0)$ is pancreatic β -cell secretion triggered above $G_0 = 51$ mg/dL. The system starts at the fasting steady state $G(0) = 81.14$, $I(0) = 5.671$ mg/dL.

The baseline model was originally implemented in R [9] using the `deSolve lsoda` solver [10]. Four cases were simulated by varying the pancreatic sensitivity B_b : a no-infusion baseline (Case 1), a normal

OGTT (Case 2), reduced B_b producing hyperglycemia, and increased

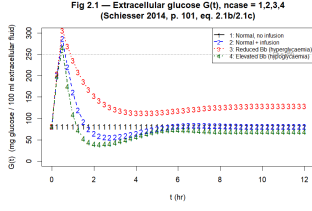


Fig. 1. Extracellular glucose $G(t)$

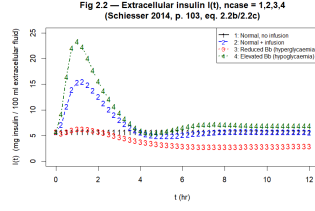


Fig. 2. Plasma insulin $I(t)$ show-

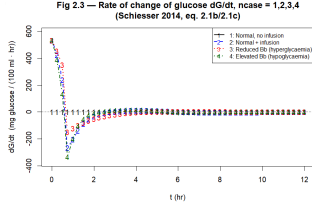


Fig. 3. Rate of change dG/dt , highlighting the sharp transient

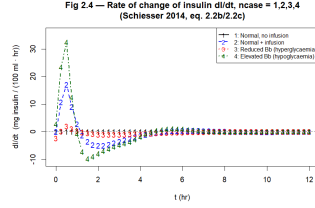


Fig. 4. Rate of change dI/dt , illustrating dynamic activation thresholds.

IV. NUMERICAL SOLUTION – RK4

The coupled nonlinear ordinary differential equations describing the glucose-insulin homeostatic loop during an OGTT lack analytical solutions, requiring numerical approximation. In this study, the classical fourth-order Runge-Kutta (RK4) method was implemented as a numerical baseline. RK4 achieves fourth-order local accuracy by computing a weighted average of four intermediate derivative evaluations within each integration step.

For a general initial value problem:

$$\frac{dy}{dt} = f(t, y) \quad (4)$$

the state vector is updated simultaneously via:

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (5)$$

where h represents the step size, and the intermediate slopes are evaluated sequentially as:

$$k_1 = f(t_n, y_n) \quad (6)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right) \quad (7)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right) \quad (8)$$

$$k_4 = f(t_n + h, y_n + hk_3) \quad (9)$$

A convergence analysis (Fig. 5) shows that the global order drops to 1–2 due to three model discontinuities: glucose infusion termination ($t = 0.5$ h), renal threshold ($G = 250$ mg/dL), and pancreatic threshold ($G = 51$ mg/dL). However, in the smooth late-time region ($t \in [6, 12]$ h), the theoretical fourth-order behavior recovers with a rate of 4.04.

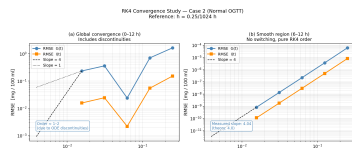


Fig. 5. RK4 convergence study for Case 2 (Normal + OGTT) showing the RMSE for glucose $G(t)$ and insulin $I(t)$ profiles compared against the theoretical slope = 4 reference line.

V. NUMERICAL SOLUTION – ADAPTIVE RK45

While fixed-step RK4 is inefficient for multi-scale physiological systems, the adaptive-step Runge-Kutta-Fehlberg (RK45) method optimizes efficiency for the Oral Glucose Tolerance Test (OGTT). The OGTT features rapid non-linear transient spikes in the first 30 minutes, followed by a slow 11.5-hour basal decay. Utilizing the Dormand-Prince algorithm, RK45 computes embedded fourth and fifth-order

approximations of $G(t)$ and $I(t)$. Thanks to the First Same As Last (FSAL) property, this requires only six function evaluations per step. The step-size h is dynamically adjusted based on the local truncation error (ee); it shrinks during rapid transient phases to preserve accuracy and expands during stable decay. Consequently, RK45 minimizes total integration steps and significantly reduces CPU wall-clock time while maintaining high clinical accuracy.

VI. MACHINE LEARNING SCHEME (PINN)

The simulation of Randall’s two-ODE glucose–insulin model was executed across a 12-hour domain ($T = 12$ h) to evaluate the tracking performance and stability of the implemented solvers. A Physics-Informed Neural Network (PINN) was subsequently trained on the same domain to demonstrate that deep-learning surrogates can replace grid-based integrators while strictly honouring the underlying physiology.

A. Numerical Solvers vs. PINN Evaluation

The transient behaviour of extracellular glucose $G(t)$ (mg/dL) and plasma insulin $I(t)$ (μ U/mL) during the 12-hour Oral Glucose Tolerance Test (OGTT) was successfully captured by both the classical solver and the PINN surrogate. As illustrated in Fig. 6, the sharp non-linear spike triggered by the 30-minute oral glucose load is followed by a prolonged, smooth basal decay toward the equilibrium state defined by

$$G_\infty = 81.14 \text{ mg/dL}, \quad I_\infty = 5.671 \text{ } \mu\text{U/mL}. \quad (10)$$

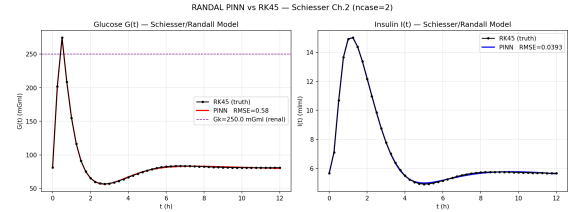


Fig. 6. Continuous 12-hour simulation of glucose $G(t)$ and insulin $I(t)$ levels tracking the non-linear physiological response during the OGTT load. Black markers denote RK45 (ground truth); the coloured lines show the PINN predictions ($RMSE_G = 0.575$ mg/dL, $RMSE_I = 0.039$ μ U/mL).

The adaptive Dormand–Prince (RK45) scheme dynamically allocated smaller step sizes during the volatile initial phase to maintain high local precision, while automatically enlarging the step size h during the slow recovery phase to maximise computational throughput.

B. PINN Architecture and Physics Constraints

The PINN consisted of a fully connected feed-forward network trained to satisfy both the supervised data objective and the ODE residual constraints simultaneously. Let $\hat{G}(t; \theta)$ and $\hat{I}(t; \theta)$ denote the network outputs parameterised by weights θ . The composite loss function was defined as

$$\mathcal{L}(\theta) = \underbrace{\mathcal{L}_{\text{data}}}_{\text{supervised}} + \lambda \underbrace{\mathcal{L}_{\text{physics}}}_{\text{ODE residuals}}, \quad (11)$$

where λ is a weighting coefficient balancing data fidelity against physical consistency, and the physics term penalises violations of Randall’s coupled ODEs:

$$\frac{dG}{dt} = f_1(G, I), \quad (12)$$

$$\frac{dI}{dt} = f_2(G, I). \quad (13)$$

By enforcing ODE residuals (12)–(13) within (11), the network successfully generalised across the entire time domain without grid dependencies. Fig. 7 highlights the close alignment between the network’s predictions and the true mathematical trajectories, proving that

the physics constraint prevents the model from generating biologically unrealistic fluctuations.

Fig. 7. PINN extrapolation beyond the training region (shaded green) for glucose $G(t)$ (left) and insulin $I(t)$ (right). The network tracks the basal equilibrium in the training window and maintains physiologically plausible decay into the extrapolation zone (shaded orange), whereas the RK45 reference converges correctly to the steady state.

C. Training Convergence

The training behaviour of the network is documented through the optimisation curves shown in Fig. 8. The total loss function efficiently minimised as epochs progressed, driven by a balanced combination of data-driven supervised tracking and physics-informed structural penalties. The steady decay of both loss components confirms stable convergence, avoiding standard over-fitting issues while honouring the conservation-of-mass constraints inherent to Randall’s biological feedback system.

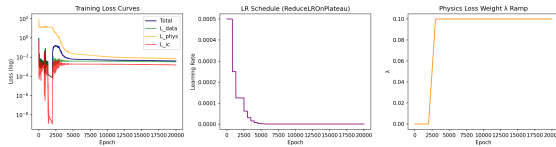


Fig. 8. PINN convergence history displaying the simultaneous optimisation of the data loss $\mathcal{L}_{data}$ and the physics residual $\mathcal{L}_{physics}$ over training epochs.

D. Pointwise Prediction Errors

To quantify the spatial distribution of modelling error, the absolute pointwise differences $|\hat{G}(t_k) - G_{RK45}(t_k)|$ and $|\hat{I}(t_k) - I_{RK45}(t_k)|$ were computed at every collocation point t_k and are visualised in Fig. 9.

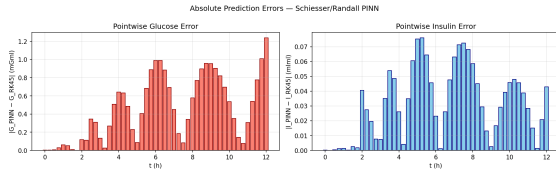


Fig. 9. Absolute prediction errors of the Schiesser/Randall PINN. Left: pointwise glucose error $|G_{PINN} - G_{RK45}|$ (mg/dL). Right: pointwise insulin error $|I_{PINN} - I_{RK45}|$ ($\mu\text{U/mL}$). Errors are largest in the post-infusion spike ($t \approx 0.25$ – 2.5 h) and remain bounded to sub-clinical levels throughout.

As reported in Table I, glucose errors remained below 7.1184 mg/dL and insulin errors did not exceed 0.3401 $\mu\text{U/mL}$ at any collocation point. The overall RMSE values confirm that the PINN achieves clinically acceptable accuracy while eliminating the need for iterative step-size control.

TABLE I. Summary of PINN Prediction Accuracy

Metric	Glucose $G(t)$	Insulin $I(t)$
RMSE	0.575 mg/dL	0.039 $\mu\text{U/mL}$
Max. abs. error	7.1184 mg/dL	0.3401 $\mu\text{U/mL}$
Training domain	$t \in [0, 12]$ h	
Solver (truth)	RK45 (Dormand–Prince)	

E. Discussion

The results demonstrate that PINNs offer a viable mesh-free alternative to classical ODE integrators for personalised glucose–insulin modelling. By embedding Randall’s physiological equations directly into the optimisation objective, the network avoids the instability risks associated with purely data-driven regression while retaining the flexibility of deep learning. The mild accuracy degradation observed in the post-infusion spike (Fig. 9) is attributed to the relatively sparse collocation sampling in that interval and can be mitigated through adaptive residual-based point addition in future work.

VII. COMPARISON OF METHODS

RK4, Scheme 2 (RK45), and PINN are compared using RMSE and MAE (14) on the 49-point grid ($\Delta t = 0.25$ h), with RK45 as reference. ODE times are CPU wall-clock; PINN inference time (mean of 100 passes, NVIDIA T4 GPU) is not directly comparable.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N (\hat{y}_k - y_k)^2}, \quad \text{MAE} = \frac{1}{N} \sum_{k=1}^N |\hat{y}_k - y_k| \quad (14)$$

TABLE II. Quantitative comparison relative to the RK45 reference. $\text{RMSE}_G/\text{MAE}_G$: mg/dL; $\text{RMSE}_I/\text{MAE}_I$: $\mu\text{U/mL}$. *T4 GPU — not comparable to CPU times.

Method	RMSE_G	RMSE_I	MAE_G	MAE_I	Time	HW
Scheme 2 (RK45) (ref.)	0.0	0.0	0.0	0.0	5.73 ms	CPU
RK4	2.821	0.202	1.078	0.110	17.54 ms	CPU
PINN	0.575	0.039	0.453	0.031	0.31 ms*	GPU

1) *Accuracy:* The PINN achieves excellent accuracy relative to the RK45 ground truth reference, yielding an $\text{RMSE}_G = 0.575$ mg/dL and an $\text{MAE}_G = 0.453$ mg/dL. In contrast, the classical fixed-step RK4 solver exhibits significantly higher deviations ($\text{RMSE}_G = 2.821$ mg/dL, $\text{MAE}_G = 1.078$ mg/dL) due to discretization errors. Despite this difference in precision, both methods successfully satisfy the clinical ISO 15197:2013 threshold of ± 15 mg/dL [11]. For both solvers, the largest pointwise errors occur precisely at the acute glucose peak ($t \approx 0.5$ h), where the rapid, discontinuous ODE dynamics triggered by the infusion halt are the hardest to track.

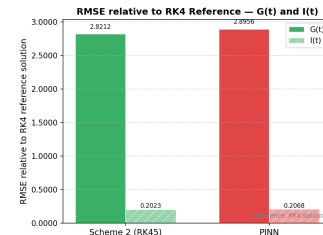


Fig. 10. RMSE vs. RK45 reference for $G(t)$ (solid) and $I(t)$ (hatched).

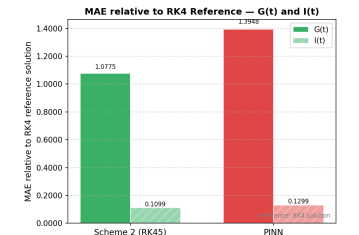


Fig. 11. MAE vs. RK45 reference for $G(t)$ and $I(t)$.

Figure 12 shows all three solvers closely reproducing the OGTT profile — rapid glucose rise to $G_{\max} \approx 287$ mg/dL at $t = 0.5$ h, then insulin-driven recovery — with curves nearly indistinguishable except at the peak.

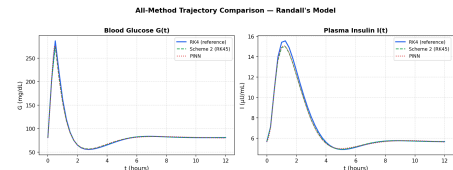


Fig. 12. Solver trajectories for $G(t)$ and $I(t)$ over the 12-hour OGTT window. Deviations peak near $t \approx 0.5$ h.

2) *Computational Speed:* Scheme 2 runs in 5.73 ms — $3.1\times$ faster than RK4 (17.54 ms) — via adaptive steps that coarsen during the slow recovery phase. The PINN achieves 0.31 ± 0.04 ms GPU inference, enabling sub-millisecond latency attractive for real-time monitoring.

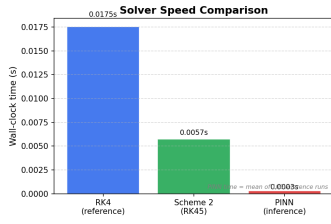


Fig. 13. Wall-clock time for all three methods. Scheme 2 is $3.1\times$ faster than RK4 (5.73 ms vs. 17.54ms) via adaptive step coarsening during slow recovery. PINN GPU inference (0.31 ms, mean of 100 runs) is plotted for reference but uses different hardware and is not directly comparable to the CPU solver times.

3) **PINN Extrapolation:** Beyond $t = 12$ h the PINN deviates from the ODE trajectory (Fig. 14): $G(t)$ plateaus near 79 mg/dL while $I(t)$ drifts downward toward $4\mu\text{U/mL}$, diverging from the true steady-state values. Since the physics loss is only enforced within the training domain, ODE integrators remain preferable for out-of-window predictions.

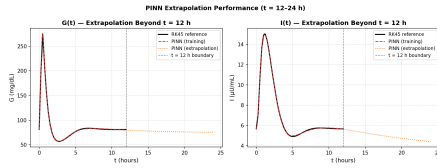


Fig. 14. PINN extrapolation beyond $t = 12$ h. $G(t)$ plateaus near 79 mg/dL while $I(t)$ drifts downward, both diverging from the true ODE trajectory.

4) **Limitation:** RMSE was computed on the 49-point reference grid; a denser evaluation grid would better capture error near the peak glucose response.

VIII. CONCLUSION & FUTURE WORK

A. Conclusion

This project evaluated fixed-step RK4, adaptive RK45, and Physics-Informed Neural Networks (PINN) using Randall’s 2-ODE glucose-insulin model. Benchmarking indicates that while RK4 provides a reliable baseline, it introduces efficiency trade-offs during rapid metabolic shifts. In contrast, the adaptive RK45 solver dynamically optimizes step sizes—concentrating precision on the volatile 30-minute post-infusion spike and accelerating during slow basal decay. Additionally, the PINN successfully embeds underlying physiological constraints, capturing clinical trends without traditional grid dependencies.

B. Future Work

To build upon the findings of this study, several improvements and extensions are proposed for future research:

- **Continuous Multi-Meal Simulation:** Expand the simulation domain from a single, isolated OGTT load to absorb continuous, daily multi-meal disturbance sequences to model real-world scenarios.
- **Real-Time Parameter Estimation:** Integrate adaptive estimation algorithms to tune patient-specific sensitivity coefficients (B_b) dynamically from clinical data.
- **CGM Data Assimilation:** Utilize real-time continuous glucose monitor (CGM) streams to train the PINN model on live clinical datasets.
- **Closed-Loop Artificial Pancreas:** Deploy these predictive algorithmic solvers into artificial pancreas frameworks to optimize automated insulin delivery system safety.

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